



5.1 Limits and the derivative from first principles

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A limit describes the value a function approaches as the input approaches some value. The derivative of f at x is defined from first principles as $f'(x) = \lim_{h \rightarrow 0} [f(x+h)-f(x)]/h$, representing the instantaneous rate of change of f , i.e. the gradient of the tangent to the curve at that point. This limit process justifies the standard differentiation rules used in practice.

Worked example: follow the reasoning

Find $f'(x)$ from first principles for $f(x)=x^2$.

$$f'(x) = \lim_{h \rightarrow 0} [(x+h)^2 - x^2]/h = \lim_{h \rightarrow 0} [x^2 + 2xh + h^2 - x^2]/h = \lim_{h \rightarrow 0} [2xh + h^2]/h = \lim_{h \rightarrow 0} (2x + h) = 2x.$$

Quick check · multiple choice

Using first principles, $f(x)=3x+2$. What is $f'(x)$?

- A 0
- B 2
- C 3
- D $3x$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Using differentiation from first principles, show that the derivative of $f(x)=x^3$ is $f'(x)=3x^2$. [5]

Final answer: _____



Answers and explanation

Question 1 [5]

$$f'(x) = \lim_{h \rightarrow 0} [(x+h)^3 - x^3]/h.$$

$$(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3.$$

$$\text{So } [(x+h)^3 - x^3]/h = [3x^2h + 3xh^2 + h^3]/h = 3x^2 + 3xh + h^2.$$

As $h \rightarrow 0$, $3xh \rightarrow 0$ and $h^2 \rightarrow 0$, so $f'(x) = 3x^2$.

Quick check answer

C. $[f(x+h) - f(x)]/h = [3(x+h) + 2 - (3x+2)]/h = 3h/h = 3$, so $f'(x) = 3$ (the constant gradient of a line).

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.